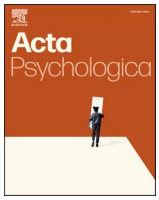
Acta Psychologica 267 (2026) 107134



Contents lists available at ScienceDirect

Acta Psychologica

journal homepage: www.elsevier.com/locate/actpsy





The relationship between ADHD, OCD, and problematic social media use

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ARTICLE INFO

Keywords:
Problematic social media use
Inattentive symptoms
Obsessive-compulsive symptoms
Social media addiction
Attention-deficit/hyperactivity disorder
University students

ABSTRACT

Problematic social media use (PSMU) has increasingly been conceptualized as an addiction-like pattern of online engagement characterized by impaired control and continued use despite negative consequences. This cross-sectional study examined whether inattentive symptoms and OCD symptoms were associated with PSMU, and whether inattentive symptoms mediated or moderated the association between OCD symptoms and PSMU, in a sample of 403 Lebanese university students who completed validated self-report measures of all three constructs. Analyses included Pearson correlations, linear regression, and PROCESS-based mediation and moderation models. Inattentive symptoms showed the strongest association with PSMU, and OCD symptoms were indirectly associated with PSMU through inattentive symptoms. Together, OCD symptoms and inattentive symptoms explained 35.98% of the variance in PSMU. Mediation analyses indicated partial mediation of the OCD–PSMU association by inattentive symptoms, whereas moderation analyses provided no evidence that inattentive symptoms altered the strength of the association between OCD symptoms and PSMU. These findings are consistent with the view that attentional dysregulation may represent an important indirect pathway linking OCD symptoms with PSMU.

1. Introduction

The marked increase in social media and other digital platforms has reshaped communication, information seeking, and social interaction, while also being accompanied by maladaptive patterns, such as compulsive checking, mood modification, withdrawal-like experiences, and social conflict, that echo key features commonly discussed within behavioral addiction frameworks (Bhukya & Lakshmana, 2025; Wartberg et al., 2025). The rapid proliferation of these platforms has therefore intensified concerns regarding mental health and well-being, particularly among adolescents and emerging adults (Baltaci, 2019; Hawi & Samaha, 2019; Hussain & Griffiths, 2018; Nguyen et al., 2020; Senturk et al., 2021). Recent research emphasizes that adolescents and students may be especially susceptible to problematic social media use (PSMU) given developmental, social, and psychological vulnerabilities (Kang et al., 2025). In addition, solitary online experiences and limited offline social engagement may further increase the risk of problematic use (Pezzi et al., 2024).

For some users, social media engagement can shift from routine use to a dysregulated pattern (i.e., PSMU). This behavior is characterized by diminished control, compulsive engagement, and persistent preoccupation with social platforms, which in turn is associated with functional impairment, distress, and social disruption (Bányai et al., 2017; Guazzini et al., 2022). Importantly, high levels of social media use do not necessarily indicate problematic use. Emerging evidence suggests that time spent on social media, in isolation, may not be inherently maladaptive, and that problematic use is better characterized by impaired control, compulsive engagement, and functional impairment. For example, recent large-scale research indicates that prolonged use may not be associated with negative outcomes when it occurs in the absence of dysregulation or distress. This distinction underscores the importance of focusing on qualitative aspects of use rather than duration alone (Zsila et al., 2026). Consistent with this profile, research associates PSMU with broader personal, interpersonal, and educational/occupational difficulties, including withdrawal-like experiences and other adverse outcomes (Cheng et al., 2021; Hawi & Samaha, 2017; Werling et al., 2022), often driven by distinct cognitive, emotional, and behavioral responses that may undermine well-being (Ansari et al., 2024). Beyond these internalizing symptoms, other psychiatric vulnerabilities have also been linked to dysregulated online social engagement (Griffiths et al., 2024).

The potentially compulsive nature of PSMU, often expressed as repetitive, habitual engagement reinforced by immediate social or informational rewards (Dolan et al., 2016; Hussain & Shabir, 2020; Whiting & Williams, 2013), has drawn comparisons to other behavioral

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https://doi.org/10.1016/j.actpsy.2026.107134
Received 21 March 2026; Received in revised form 7 May 2026; Accepted 22 May 2026
Available online 25 May 2026
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addictions (D'Arienzo et al., 2019), highlighting the need to clarify its psychological and clinical correlates. One vulnerability that has received increasing attention is attention-deficit/hyperactivity disorder (ADHD). ADHD is a neurodevelopmental condition characterized by developmentally inappropriate patterns of inattention and/or hyperactivity–impulsivity that interfere with functioning. Evidence suggests that ADHD symptoms are associated with higher susceptibility to PSMU, with adolescents and emerging adults showing heightened emotional and behavioral difficulties that may amplify problematic engagement (Dekkers & van Hoorn, 2022; Shuai et al., 2021).

In particular, inattention may be associated with increased vulnerability by undermining sustained attention, task persistence, and self-regulation, which may be associated with increased susceptibility to compulsive checking and difficulty disengaging from rewarding digital environments (Dekkers & van Hoorn, 2022). Consistent with this account, multiple studies indicate that inattention is among the strongest predictors of PSMU, associating it with prolonged screen time, difficulties disengaging from platforms, and excessive online multitasking (Boer et al., 2020; Deasley, 2016; Settanni et al., 2018; Thorell et al., 2024). Obsessive-compulsive disorder (OCD) may add another layer of complexity to understanding PSMU. Characterized by intrusive thoughts (obsessions) and repetitive behaviors (compulsions) intended to reduce distress, OCD shares behavioral parallels with PSMU, including compulsive checking, reassurance-seeking, and difficulties regulating use (Guazzini et al., 2022; van Bennekom et al., 2018). Social media platforms, through rapid feedback and intermittent reinforcement, may further maintain these patterns and may be associated with greater vulnerability to PSMU among individuals with elevated anxiety and dependence-prone traits (James et al., 2017; Senturk et al., 2021). Fear of missing out (FOMO) may also intensify repetitive engagement, potentially contributing to fatigue and burnout among individuals with OCD (Fontes-Perryman & Spina, 2022).

From this perspective, social media platforms are not viewed as inherently addictive. Rather, problematic use may emerge when these platforms are used to manage underlying psychological distress, thereby reinforcing maladaptive engagement over time (Griffiths, 2005; Hussain & Griffiths, 2018). Within cognitive-behavioral accounts of problematic online behavior, psychopathological vulnerabilities may function as distal risk factors that shape maladaptive cognitions and reinforce dysregulated patterns of use (Gámez-Guadix et al., 2013, 2012). In this context, inattentive symptoms and OCD symptoms may be especially relevant. Inattention may be associated with greater vulnerability to PSMU through difficulties in sustained attention, self-regulation, and disengagement from salient digital cues (Arness & Ollis, 2023; Dekkers & van Hoorn, 2022; Liu et al., 2024), whereas OCD symptoms may be associated with repetitive checking, reassurance-seeking, and compulsive patterns of online engagement (Fontes-Perryman & Spina, 2022; van Bennekom et al., 2018).

Despite growing evidence linking both inattentive symptoms and OCD symptoms to problematic online behaviors (Arness & Ollis, 2023; Fontes-Perryman & Spina, 2022; Liu et al., 2024), the nature of their joint association with PSMU remains insufficiently understood. In particular, it remains unclear whether inattentive symptoms operate primarily as an indirect pathway through which OCD symptoms are associated with PSMU, as a vulnerability factor that changes the strength of that association, or both. Addressing this question may help clarify the cognitive and clinical correlates of PSMU among university students (Awad et al., 2022; Fekih-Romdhane et al., 2024). University students represent a particularly relevant population for examining problematic social media use, as emerging adulthood is characterized by increased autonomy, academic demands, and heightened reliance on digital communication (Nguyen et al., 2020; Whiting & Williams, 2013). In the Lebanese context, where social media platforms play an important role in communication and information exchange, and maintaining social connections, these behaviors are highly integrated into students' daily lives (Awad et al., 2022; Fekih-Romdhane et al., 2024). Prior research among Lebanese university students has also reported elevated technology-related addictive behaviors, including excessive social media use (Samaha & Hawi, 2016), and has linked these behaviors with psychological and behavioral difficulties (Sabbah et al., 2019). Accordingly, examining the psychological correlates of PSMU in Lebanese university students may contribute to a more contextually grounded understanding of problematic digital engagement.

Accordingly, the present study examined the association between inattentive symptoms, OCD symptoms, and PSMU among Lebanese university students. First, it was expected that higher levels of inattentive symptoms would be positively associated with PSMU (H1). Second, it was hypothesized that inattentive symptoms would partially mediate the association between OCD symptoms and PSMU, such that higher OCD symptom severity would be associated with greater inattentive symptoms, which in turn would be associated with higher PSMU (H2). Third, it was hypothesized that inattentive symptoms would moderate the association between OCD symptoms and PSMU, such that the association between OCD symptoms and PSMU would be stronger at higher levels of inattention (H3).

To test these hypotheses, mediation and moderation analyses were conducted using the PROCESS macro for SPSS. In doing so, the study examined whether inattentive symptoms functioned more strongly as an indirect pathway or as a moderator of the association between OCD symptoms and PSMU. Clarifying these relationships may contribute to a more precise understanding of the psychological factors associated with problematic social media use in student populations.

2. Method

2.1. Participants and procedure

The present study employed a cross-sectional, online survey design to examine the relationships among OCD symptoms, ADHD symptoms, and PSMU among a sample of Lebanese university students. Approval for the study was obtained from the university institutional review board (IRB Reference number: IRB 202428). Participants were recruited via convenience sampling using mailing lists and social media platforms between December 2024 and January 2025. Before starting the survey, participants were given an overview of the study's purpose and assurances of confidentiality and anonymity regarding data collection, storage, and reporting. All participants provided written informed consent to participate in the study before being given access to the survey.

A total of 403 undergraduate students from a private university completed the survey ($M_{age} = 20.11$ years, $SD = 1.83$), with ages ranging from 18 to 26 years. The sample included students from sophomore to senior year. The sample size was determined using an a priori power analysis conducted in *G*Power* 3.1 for multiple linear regression. Assuming a medium effect size ($f^2 = 0.15$), $\alpha = 0.05$, power = 0.95, and 3 predictors, the minimum required sample size was 119. The final sample of 403 participants exceeded this requirement. Of the 403 participants, 181 were female (44.9%) and 222 were male (55.1%). Participants were recruited from a range of broad academic disciplines. Typically, participants completed the survey within 25 min. No missing data were observed for age or gender.

2.2. Measures

The survey comprised four sections: demographic questions, PSMU assessment, OCD assessment, and ADHD assessment. The demographic section collected data on age and gender.

PSMU was assessed using the eight-item Social Media Addiction Questionnaire (SMAQ; Hawi & Samaha, 2017). Items (e.g., "I have been unable to reduce my use of social media") are rated on a 7-point Likert scale ranging from 1 (strongly disagree) to 7 (strongly agree). Following the standard SMAQ scoring procedure, items endorsed at the threshold level (ratings of 6 or 7) were coded as 1, with Items 1 and 2 contributing

a maximum combined score of 1 and Items 4 and 5 contributing a maximum combined score of 1, yielding a total score range of 0 to 6. In the present study, PSMU was operationalized as this threshold-scored SMAQ total, with higher scores indicating greater problematic social media use. Although a total score of 5 or 6 is used to indicate probable social media addiction, this classification was used only for descriptive purposes. The continuous SMAQ total score was used in all correlational, regression, mediation, and moderation analyses. The SMAQ demonstrated very good internal consistency in the present study (Cronbach's $\alpha = 0.851$; McDonald's $\omega = 0.844$).

ADHD symptoms were assessed using the 18-item Adult ADHD Self-Report Scale (ASRS-v1.1), developed by the World Health Organization (WHO) based on DSM-IV criteria (American Psychiatric Association, 2000; Kessler et al., 2005). The ASRS assesses core ADHD symptoms, specifically inattention and hyperactivity/impulsivity, among adult populations (Kessler et al., 2007). Although the scale is not intended for diagnostic purposes, the ASRS serves as a validated screening instrument for identifying individuals whose symptom patterns may warrant further clinical evaluation. The ASRS-v1.1 comprises two subscales: the Inattention subscale and the Hyperactivity/Impulsivity subscale. The nine-item Inattention subscale assesses inattentive symptoms (i.e., self-reported attentional difficulties), including difficulties related to sustained attention, organization, and concentration. Items (e.g., "*How often are you distracted by activity or noise around you?*") are scored dichotomously (0 or 1), with total scores ranging from 0 to 9. The nine-item Hyperactivity/Impulsivity subscale assesses motor symptoms (five items) and verbal impulsivity (four items). Items (e.g., "*How often do you interrupt others when they are busy?*") are also scored dichotomously (0 or 1), with total scores ranging from 0 to 9. Scoring followed the standard ASRS scoring procedure, whereby each item was assigned either 0 or 1. For Items 1–3, 9, 12, 16, and 18, responses of "sometimes", "often," or "very often" were scored as 1, whereas "never" or "rarely" were scored as 0. For the remaining items, only "often" or "very often" were scored as 1, and "never", "rarely" or "sometimes" were scored as 0. In the present study, the ASRS Inattention subscale was used to assess inattentive symptoms dimensionally in a non-clinical sample and was analyzed as a continuous symptom variable. Although ASRS subscale scores can be used for screening purposes, screening classifications were not used in the inferential analyses. The ASRS-v1.1 demonstrated very good internal consistency in the present study (Cronbach's $\alpha = 0.855$; McDonald's $\omega = 0.852$).

OCD symptoms were assessed using the 18-item Obsessive-Compulsive Inventory-Revised (OCI-R) (Foa et al., 2002). The scale comprises six dimensions common to OCD: checking, ordering, neutralizing, washing, obsessing, and hoarding. Items (e.g., *"I check things more often than necessary"*) are rated on a five-point Likert scale from 0 (*Not at all*) to 4 (*Extremely*), with total scores ranging from 0 to 72. In the present study, OCD symptoms were analyzed as a continuous total OCI-R score, with higher scores indicating greater obsessive-compulsive symptom severity. Although a cutoff score has been proposed in prior research for screening purposes, the continuous total score was used in all inferential analyses. The OCI-R demonstrated very good internal consistency in the present study (Cronbach's $\alpha = 0.884$; McDonald's $\omega = 0.881$).

2.3. Data analysis

Data analysis was conducted using SPSS (Version 27), except for McDonald's omega, which was calculated using R (Version 4.4.2) and the psych package to assess the internal consistency of the OCI-R, ASRS-v1.1, and SMAQ. Descriptive statistics were computed for demographic characteristics and study variables. Pearson's correlation coefficients were used to assess the associations among OCD symptoms, inattentive symptoms, and PSMU. All inferential analyses treated PSMU as a continuous outcome variable, operationalized as the threshold-scored SMAQ total score (range = 0–6), and treated inattentive symptoms as a continuous variable, operationalized as the ASRS Inattention subscale total score. OCD symptoms were operationalized as the continuous total OCI-R score. No demographic covariates were included in the inferential models.

A simple linear regression analysis was conducted to examine whether inattentive symptoms were associated with PSMU. Mediation and moderation analyses were conducted using the PROCESS macro for SPSS (Version 5.0). PROCESS Model 4 was used to test whether inattentive symptoms mediated the association between OCD symptoms and PSMU. PROCESS Model 1 was used to test whether inattentive symptoms moderated the association between OCD symptoms and PSMU by including the product term between OCD symptoms and inattentive symptoms. For the moderation analysis, conditional effects were examined at low, moderate, and high levels of inattentive symptoms (values of 1.0, 4.0, and 6.0, respectively). Indirect effects were estimated using bias-corrected bootstrapping with 5000 resamples, and statistical significance was determined using 95% confidence intervals that did not include zero. Unstandardized regression coefficients, standard errors, *p* values, and confidence intervals were reported for the regression-based analyses.

Prior to conducting the regression-based analyses, assumptions relevant to linear regression were evaluated. Linearity between the predictor variables and PSMU was assessed through visual inspection of scatterplots. Homoscedasticity was examined by inspecting plots of standardized residuals against standardized predicted values. Normality was evaluated by examining the distribution of standardized residuals and normal probability plots. Multicollinearity between OCD symptoms and inattentive symptoms was assessed using tolerance and variance inflation factor values. Because mediation effects were estimated using bias-corrected bootstrapping with 5000 resamples, inference for the indirect effect did not rely on the assumption of normality of the sampling distribution of the indirect effect. Missing data were handled using listwise deletion, with participants who had missing values on any study variable excluded from the relevant analyses.

3. Results

3.1. Descriptive statistics and screening-based classifications

The descriptive statistics for the study variables, including means, standard deviations, observed ranges, and the prevalence of participants meeting screening thresholds, are presented in Table 1. The mean SMAQ total score was 2.22 (*SD* = 1.30, observed range = 1–6), the mean OCI-R total score was 25.43 (*SD* = 11.97, observed range = 2–64), and the mean ASRS Inattention subscale score was 3.60 (*SD* = 2.26, observed

Table 1
Descriptive statistics and screening-based classifications for ADHD, OCD, and PSMU.

Variable	*M*	*SD*	Observed range	Cutoff for screening	% Above cutoff
SMAQ total score (0–6)	2.22	1.30	1–6	≥5	8.00
OCI-R total score (0–72)	25.43	11.97	2–64	≥28	36.70
ASRS Inattention subscale (0–9)	3.60	2.26	0–9	≥5	28.78
ASRS Hyperactivity/Impulsivity subscale (0–9)	2.75	2.21	0–9	≥5	20.54
ASRS Total score (0–18)	6.40	4.11	0–17	≥9	29.31

Note. SMAQ = Social Media Addiction Questionnaire; OCI-R = Obsessive-Compulsive Inventory–Revised; ASRS = Adult ADHD Self-Report Scale. Means, standard deviations, and observed ranges are reported for continuous variables. Screening classifications are shown for descriptive purposes only and were not used in the inferential analyses.

range = 0–9). For descriptive purposes, 36.70% of participants screened positive for OCD symptoms (OCI-$R \geq 28$), 29.31% screened positive for ADHD symptoms (ASRS Total score ≥ 9), and 8.00% met the cutoff for probable PSMU (SMAQ ≥ 5). Scores on the ASRS Inattention subscale ($M = 3.60$, $SD = 2.26$) were higher than scores on the ASRS Hyperactivity/Impulsivity subscale ($M = 2.75$, $SD = 2.21$). In addition, 28.78% of participants met the screening threshold for inattention (≥ 5), compared with 20.54% for hyperactivity/impulsivity (≥ 5).

Pearson product-moment correlations among the continuous study variables are presented in Table 2. PSMU was positively correlated with OCD symptoms and with overall ADHD symptoms and all ADHD subscales. Among the ADHD-related variables, the strongest correlation with PSMU was observed for inattentive symptoms. OCD symptoms were also positively correlated with overall ADHD symptoms and all ADHD subscales, with the strongest association observed for inattentive symptoms.

Although inattentive symptoms showed the strongest correlation with PSMU, motor and verbal impulsivity symptoms were also positively associated with PSMU. This indicates that broader ADHD-related symptom dimensions may also be relevant.

3.2. Regression analysis

Given the strongest correlation between PSMU and inattentive symptoms in Table 2, a simple linear regression analysis was conducted to examine whether inattentive symptoms were associated with PSMU. The model was statistically significant, $F(1, 401) = 216.509$, $p < .001$, explaining 35.1% of the variance in PSMU ($R^2 = 0.351$, adjusted $R^2 = 0.349$). Higher inattention scores were associated with higher PSMU scores, $B = 0.428$, $SE = 0.029$, $\beta = 0.592$, $t = 14.714$, $p < .001$ (see Table 3).

3.3. Mediation analysis

Mediation analysis using the PROCESS macro (Model 4) examined whether inattentive symptoms mediated the association between OCD symptoms and PSMU (Table 4). OCD symptoms were significantly associated with inattentive symptoms ($B = 0.0691$, $p < .001$, 95% *CI* [0.0533, 0.0849]), accounting for 15.56% of the variance in inattentive symptoms ($R^2 = 0.1556$). In the outcome model, OCD symptoms and inattentive symptoms jointly explained 35.98% of the variance in PSMU ($R^2 = 0.3598$).

The direct effect of OCD symptoms on PSMU was statistically significant but small ($B = 0.0130$, $p = .0188$). In addition, inattentive symptoms were significantly associated with PSMU ($B = 0.3987$, $p < .001$), and the indirect effect of OCD symptoms on PSMU through inattentive symptoms was significant ($B = 0.0276$, 95% *CI* [0.0194, 0.0358]), indicating partial mediation (Fig. 1). The total effect of OCD symptoms on PSMU was 0.0406, with approximately 68% of the total effect represented by the indirect pathway through inattention. Overall, this pattern is consistent with a statistically significant indirect association between OCD symptoms and PSMU through inattentive symptoms.

Multicollinearity diagnostics for the outcome model indicated no evidence of problematic collinearity between OCD symptoms and inattentive symptoms, with VIF values of 1.21 and tolerance values of 0.83 for both predictors. To further examine the robustness of the mediation findings to demographic variation, we re-estimated the mediation model while including gender as a covariate in both the mediator and outcome equations. The adjusted model showed that OCD symptoms remained significantly associated with inattentive symptoms ($b = 0.071$, 95% bootstrap *CI* [0.056, 0.087], $p < .001$), and inattentive symptoms remained significantly associated with problematic social media use ($b = 0.316$, 95% bootstrap *CI* [0.243, 0.383], $p < .001$). The indirect effect of OCD symptoms on problematic social media use through inattentive symptoms remained significant after adjustment for gender ($b = 0.023$, 95% bootstrap *CI* [0.016, 0.030], $p < .001$), while the direct effect remained small but statistically significant ($b = 0.012$, 95% bootstrap *CI* [0.001, 0.024], $p = .043$). Gender was not a significant covariate for inattentive symptoms ($b = 0.066$, $p = .784$) or problematic social media use ($b = 0.126$, $p = .335$). These findings indicate that the observed mediation pattern was robust to adjustment for gender.

3.4. Moderation analysis

The moderation analysis examined whether inattentive symptoms moderated the relationship between OCD symptoms and PSMU (Table 5). The overall model was significant ($R^2 = 0.3598$, $p < .001$), indicating that approximately 36% of the variance in PSMU was explained by the predictors. Consistent with prior analyses, inattentive symptoms showed a significant association with PSMU ($B = 0.3772$, $p < .001$). However, neither the main effect of OCD symptoms ($B = 0.0096$, $p = .3406$) nor the interaction effect (OCD × inattentive symptoms) was significant ($B = 0.0009$, $p = .6871$). The non-significant interaction indicated that inattentive symptoms did not moderate the OCD-PSMU relationship (i.e., the association between OCD symptoms and PSMU did not vary across levels of inattention). Taken together, these findings indicate that while inattentive symptoms were a significant predictor of PSMU, they did not alter the strength of the association between OCD symptoms and PSMU. Thus, the moderation hypothesis was not supported.

Table 2
Pearson product-moment correlation coefficients between research variables.

Variable	ADHD	Inattention	Motor	Verbal	PSMU
OCD	0.299**	0.394**	0.316**	0.298**	0.320**
ADHD		0.752**	0.664**	0.494**	0.439**
Inattention			0.463**	0.498**	0.592**
Motor				0.576**	0.358**
Verbal					0.375**

Note. PSMU = problematic social media use; OCD = obsessive-compulsive symptoms; ADHD = attention-deficit/hyperactivity symptoms; Motor = ADHD motor subscale; Verbal = ADHD verbal impulsivity subscale; Inattention = ADHD inattention subscale. *$p < .05$, **$p < .01$.

Table 3
Linear regression analysis predicting PSMU from inattentive symptoms.

Predictor	*B*	*SE B*	*β*	*t*	*p*	*95% CI*
Inattention	0.428	0.029	0.592	14.714	<0.001	[0.371, 0.485]
Intercept	0.753	0.123	–	6.109	<0.001	[0.512, 0.994]

Note. PSMU = problematic social media use; ASRS = Adult ADHD Self-Report Scale. Model summary: $R^2 = 0.351$, adjusted $R^2 = 0.349$, $F(1, 401) = 216.509$, $p < .001$.

Table 4
Mediation analysis of inattentive symptoms in the association between OCD symptoms and PSMU.

Path	*B*	*SE*	*t*	95% *CI*
OCD → Inattentive ADHD (Path a)	0.0691	0.0080	8.5946	[0.0533, 0.0849]
Inattentive symptoms → PSMU (Path b)	0.3987	0.0315	12.6681	[0.3368, 0.4606]
OCD → PSMU (Direct effect, c')	0.0130	0.0055	2.3582	[0.0022, 0.0239]
OCD → Inattentive symptoms → PSMU (Indirect effect, ab)	0.0276	0.0042	–	[0.0194, 0.0358]

Note. PSMU = problematic social media use; OCD = obsessive-compulsive symptoms. Indirect effects were estimated using bias-corrected bootstrapping with 5000 resamples. The indirect effect is statistically significant when the 95% confidence interval does not include zero.

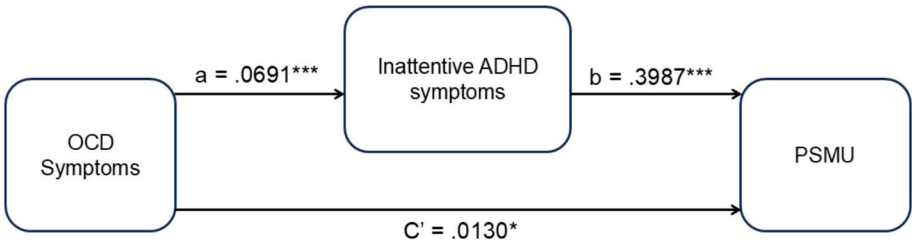


Fig. 1. Mediation Model of OCD → Inattention → PSMU.
Note. Unstandardized coefficients are reported; Indirect effect = 0.0276 (95% *CI* [0.0194, 0.0358]).
*$p < .05$, ***$p < .001$.

Table 5
Moderation analysis of inattentive symptoms in the association between OCD symptoms and PSMU.

Effect	*B*	*SE*	*t*	*p*	*95% CI*
OCD	0.0096	0.0101	0.9541	0.3406	[−0.0102, 0.0294]
Inattentive symptoms	0.3772	0.0619	6.0934	<0.001	[0.2555, 0.4989]
OCD x Inattentive symptoms	0.0009	0.0021	0.4031	0.6871	[−0.0033, 0.0050]

Note. PSMU = problematic social media use; OCD = obsessive-compulsive symptoms. The interaction term tested whether inattentive symptoms moderated the association between OCD symptoms and PSMU. Model summary. $R^2 = 0.3598$, $F(3, 399) = 74.73$, $p < .001$.

4. Discussion

The present study examined the associations among inattentive symptoms, obsessive-compulsive symptoms, and problematic social media use (PSMU) in a sample of Lebanese university students. Three main findings emerged. First, inattentive symptoms were strongly associated with PSMU. Second, inattentive symptoms partially mediated the association between OCD symptoms and PSMU. Third, no evidence was found for a moderating effect of inattentive symptoms on the OCD–PSMU association. Taken together, these findings are most consistent with a partial mediation model in which attentional dysregulation may represent a potential pathway linking obsessive-compulsive symptomatology with problematic social media use.

These findings suggest that the association between OCD symptoms and problematic social media use may operate, in part, through attentional processes. Individuals with higher levels of obsessive-compulsive symptoms may experience greater difficulties in attentional control and disengagement, which may be associated with increased vulnerability to repetitive and cue-driven patterns of social media engagement. In contrast, the absence of a moderating effect indicates that inattention does not amplify the strength of the OCD-PSMU association across different levels of symptom severity but may instead function as a pathway through which this association is expressed.

The present findings also have tentative applied implications. Attentional difficulties and obsessive-compulsive tendencies may be associated with increased vulnerability to problematic social media use, suggesting that interventions targeting attentional regulation, cognitive flexibility, and compulsivity-related processes may be relevant. Approaches such as cognitive-behavioral and mindfulness-based interventions, along with digital well-being strategies (e.g., structured screen-time management), may therefore be worth examining in future research. In addition to reducing problematic use, emerging research highlights the importance of promoting adaptive forms of social media engagement, such as mindful use. Mindful engagement emphasizes intentional, present-focused interaction with digital platforms and has been associated with improved well-being and lower levels of compulsive use patterns (Shabahang et al., 2024). These approaches may complement interventions focused solely on reducing usage. However, these implications should be interpreted cautiously, as the present study did not directly examine intervention effects.

Inattentive symptoms showed the strongest association with PSMU, supporting H1. However, the significant correlations between PSMU and the motor and verbal impulsivity subscales indicate that impulsivity-related symptoms may also contribute to problematic social media use. These associations may reflect difficulties inhibiting urges to check social media, respond to notifications, or disengage from immediately rewarding online cues. Therefore, the present findings should not be interpreted as excluding the role of impulsivity. Rather, future research should examine inattentive and impulsivity-related symptoms simultaneously to clarify their unique and shared contributions to PSMU. Consistent with prior research, the strong association between inattention and PSMU may reflect difficulties in self-regulation, sustained attention, and disengagement from salient digital cues (Arness & Ollis, 2023; Werling et al., 2022). The pattern of findings also suggests that the association between ADHD-related symptoms and PSMU may be more strongly linked to inattention-related processes than to broader ADHD symptomatology.

The observed association between inattentive symptoms and PSMU also provides context for the mediation findings involving OCD symptoms. Supporting H2, inattentive symptoms partially mediated the association between OCD symptoms and PSMU, suggesting that attentional processes may play a role in linking obsessive-compulsive symptomatology with problematic social media use. This interpretation is consistent with executive dysfunction accounts, which propose that individuals with elevated OCD symptoms may experience difficulties in cognitive control, attentional disengagement, and the regulation of repetitive or habit-driven responses. In digital contexts, such difficulties may contribute to more persistent checking and greater difficulty disengaging from reinforcing social media cues, which may in turn be associated with PSMU (Abramovitch et al., 2019).

These supplementary analyses strengthen confidence in the robustness of the mediation findings. First, the low VIF and high tolerance values indicate that the observed associations were not attributable to problematic multicollinearity between OCD symptoms and inattentive symptoms. Second, the indirect pathway from OCD symptoms to problematic social media use through inattentive symptoms remained significant after adjusting for gender, while gender itself was not a significant covariate in the model. This pattern suggests that the mediating role of inattentive symptoms is substantive rather than an artifact of demographic variation. The subgroup analyses were consistent with this interpretation, as the indirect effect was evident in both males and females, although the residual direct association appeared more pronounced in males.

In the present study, approximately 68% of the total association between OCD symptoms and PSMU was represented by the mediated association through inattentive symptoms. These findings suggest that attentional processes may account for a substantial portion of this association. This pattern is consistent with the possibility that difficulties in sustained attention and disengagement are associated with greater

vulnerability to repetitive, cue-driven online behaviors, which may be associated with greater difficulty regulating problematic social media use. In turn, problematic online engagement may become negatively reinforced as a maladaptive coping response to distress and may contribute to the maintenance of PSMU over time (Choudhury & Ali, 2020; Ioannidis et al., 2018; Thorell et al., 2024). By modeling inattentive symptoms as a mediator, the present study offers a more nuanced perspective on the associations among OCD symptoms, inattention, and PSMU. However, contrary to H3, inattentive symptoms did not moderate the association between OCD symptoms and PSMU, indicating that the strength of this relationship did not vary significantly across levels of inattention. Thus, although inattention was strongly associated with PSMU, it did not function as a risk amplifier of the association between OCD symptoms and PSMU.

In addition to the mediation findings, the regression-based analyses indicated that inattentive symptoms were consistently associated with PSMU across models. The proportion of explained variance ranged from 15.56% in the model predicting inattentive symptoms to 35.98% in the model predicting PSMU from OCD symptoms and inattentive symptoms. Taken together, these findings suggest that inattention is more relevant to the OCD-PSMU association as a mediator than as a moderator. More specifically, OCD symptoms showed a small but significant direct association with PSMU, while a larger portion of the overall association was represented by the indirect association through inattentive symptoms. This pattern is broadly consistent with prior research linking obsessive-compulsive symptoms, attentional difficulties, and problematic online behaviors, including findings relating OCD symptoms to fear of missing out and digital fatigue, and ADHD symptoms to emotional dysregulation and addictive online behaviors (Fontes-Perryman & Spina, 2022; Liu et al., 2024).

Several limitations should be considered when interpreting the present findings. First, the sample consisted of Lebanese university students from a private university, which may limit the generalizability of the results to other cultural, demographic, or educational contexts. Second, all variables were assessed using self-report measures, which may be influenced by recall bias and social desirability. Although gender was collected as a demographic variable, the study did not assess sexual orientation, gender identity beyond the available response categories, or LGBTQ status. This limits the ability to examine whether patterns of PSMU differ across diverse gender and sexual identity groups. Future research should include more inclusive demographic measures to better understand these potentially important differences.

Third, OCD and inattentive symptoms were operationalized as continuous dimensions rather than clinical diagnoses, which constrains the applicability of the findings to clinical populations. Fourth, demographic covariates were not initially included in the inferential models, and thus the possibility of residual confounding cannot be ruled out. Finally, the cross-sectional design precludes causal inference and limits conclusions regarding the directionality of the observed associations.

Future research should adopt longitudinal and multi-method approaches to clarify the temporal ordering and underlying mechanisms linking OCD symptoms, inattentive symptoms, and problematic social media use (PSMU). Additionally, examining contextual moderators, such as peer influences, platform-specific features, and broader aspects of the digital environment, may provide a more nuanced understanding of these relationships, particularly within emerging adult populations. Incorporating comorbid conditions, including anxiety and depression, may further enhance understanding of the broader psychological context in which these variables interact.

5. Conclusion

The present study suggests that inattentive symptoms may represent a theoretically informed indirect pathway statistically linking OCD symptoms with problematic social media use (PSMU), with approximately 68% of the total association between OCD symptoms and PSMU represented by the indirect pathway through inattention. Overall, the findings are consistent with the view that OCD symptoms are associated with PSMU partly through attentional dysregulation, alongside a smaller direct association. Contrary to H3, inattentive symptoms did not moderate this relationship, indicating that the association between OCD symptoms and PSMU did not vary significantly across levels of inattention. Taken together, these findings highlight the potential relevance of attentional regulation and compulsivity-related processes in understanding problematic social media use. Future research should use longitudinal and multi-method designs to clarify temporal ordering and causal direction, and should also examine the roles of psychiatric comorbidity and demographic factors in shaping these associations.

CRediT authorship contribution statement

Nazir S. Hawi: Formal analysis. **Maya Samaha:** Conceptualization. **Emma Shaffu:** Data curation.

Consent to participate

Informed consent was obtained from all individual participants included in the study.

Ethical approval

Approval for this study including the questionnaire and methodology was obtained by the University Institutional Review Board at Notre Dame University-Louaize IRB Reference #: IRB 202428.

Funding

Not applicable.

Declaration of competing interest

On behalf of all authors, the corresponding author states that there is no conflict of interest.

Data availability

Data will be made available on request.

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